The Arrow of Time as Petz Recovery Failure

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We define a temporal asymmetry parameter $\tau = 1 - F$, where F is the fidelity of Petz recovery, and argue that it provides a continuous, operational measure of the arrow of time: $\tau = 0$ for closed, unitary evolution and $\tau > 0$ for irreversible processes, bounded by entropy production via $\tau \leq 1 - \exp(-\Sigma/2)$ (restating the universal recovery bound of Junge et al.). Because the Petz map is the unique retrodiction functor satisfying Bayesian consistency, this single parameter links the quantum eraser (where $\tau = 0$ because the signal-idler pair is closed), quantum error correction (where $\tau > 0$ sets the decoder's task), and thermodynamic irreversibility (where τ is bounded by dissipation). We establish two new results: the recovery bound $F^2 \geq \exp(-\Delta D)$ is exactly saturated if and only if the channel acts as a quantum sufficient statistic (proved analytically for commuting outputs, verified numerically for the non-commuting case), and $\sqrt{\tau}$ satisfies a triangle inequality under channel composition—a property unique to the standard Petz map.

INTRODUCTION

The Petz recovery map [3, 4] is a near-optimal quantum error correction (QEC) decoder [5], the *unique* retrodiction functor satisfying Bayesian consistency [6], and the reverse channel in quantum Crooks theorems [7]. It also appears in entanglement wedge reconstruction [8] and as a physical reversal via dilations [17]. Concurrent work by Bai, Buscemi, and Scarani [31] develops stochastic entropy production using the same Petz-retrodiction foundation.

In this Letter, we offer a physical interpretation of this mathematical unity. We define a temporal asymmetry parameter $\tau = 1 - F$, where F is the Petz recovery fidelity, and argue that τ provides a continuous, operational measure of the arrow of time.

The interpretive framework rests on known results assembled into a single picture. The quantum eraser [1, 2] is a physical instantiation of Petz recovery, and the uniqueness of the Petz map as retrodiction functor (due to Parzygnat–Buscemi [6]) explains why it also serves as a near-optimal QEC decoder [5, 11]. The universal recovery bound of Junge et al. [11] (Corollary 5.1 therein) and the Fawzi–Renner inequality [12], combined with the Kolchinsky–Wolpert decomposition [13] and the Reeb–Wolf bound [18], yield a chain linking recovery fidelity to entropy production. The logical flow is:

$$\text{Eraser} \xrightarrow{\text{Petz}} \text{Retrodiction} \to \text{QEC} \to \text{Thermo}. \tag{1}$$

The physical content—that Petz recovery failure quantifies the arrow of time—is made precise in Sec. .

We present two new mathematical results. First, we characterize exactly when the Junge et al. bound $F^2 \geq \exp(-\Delta D)$ is saturated: if and only if the channel acts as a quantum sufficient statistic (Theorem 3). Second, we prove that $\sqrt{\tau}$ satisfies a triangle inequality under channel composition—a structural property unique to the Petz map that fails for the rotated Petz map (Theorem 4).

THE PETZ RECOVERY MAP

Given a quantum channel $\mathcal{N} : \mathcal{B}(\mathcal{H}_A) \to \mathcal{B}(\mathcal{H}_B)$ and a faithful reference state $\sigma \in \mathcal{S}(\mathcal{H}_A)$, the *Petz recovery map* is

$$\mathcal{R}_{\sigma,\mathcal{N}}(\rho) = \sigma^{1/2}\,\mathcal{N}^\dagger\big(\mathcal{N}(\sigma)^{-1/2}\,\rho\,\mathcal{N}(\sigma)^{-1/2}\big)\,\sigma^{1/2}, \tag{2}$$

where $\mathcal{N}^\dagger$ is the Hilbert–Schmidt adjoint of $\mathcal{N}$. The map $\mathcal{R}_{\sigma,\mathcal{N}}$ is completely positive and trace-preserving (CPTP) [3], and satisfies exact recovery $\mathcal{R}_{\sigma,\mathcal{N}} \circ \mathcal{N} = \mathrm{id}$ on the support of σ if and only if the data-processing inequality (DPI) for relative entropy is saturated: $D(\sigma\|\tau) = D(\mathcal{N}(\sigma)\|\mathcal{N}(\tau))$ for all τ [4].

Table I shows how the mathematical ingredients of Eq. (2) map to physical meaning in each domain.

TABLE I. Physical interpretation of the Petz map $\mathcal{R}_{\sigma,\mathcal{N}}$ across four domains.

	Eraser	**Time Rev.**	**QEC**	**Thermo.**
$\mathcal{N}$	Path mark	Fwd. evol.	Noise	Irrev. proc.
σ	Entangled	Prior	Code state	Gibbs
$\mathcal{R}$	Erasure	Retrodiction	Decoder	Rev. proc.
ΔD	0 (closed)	Info. loss	Code dist.	βW_{diss}

UNIQUENESS OF RETRODICTION

We recall the uniqueness result of Parzygnat and Buscemi [6], which is central to our interpretation.

Theorem 1 (Retrodiction Uniqueness (Parzygnat–Buscemi [6])). *Let* C *be the category of finite-dimensional* C^**-algebras with state-preserving CP maps as morphisms. A* retrodiction functor *is a contravariant functor* $R : \mathsf{C} \to \mathsf{C}^{\mathrm{op}}$ *satisfying:*

(i) Preservation: *R sends $(\mathcal{N}, \sigma)$ to a CPTP map.*

(ii) Bayesian consistency: *For composable channels* $\mathcal{N}_1, \mathcal{N}_2$ *and prior* σ*,*

$$R(\mathcal{N}_2 \circ \mathcal{N}_1,\, \sigma) = R(\mathcal{N}_1,\, \sigma) \circ R(\mathcal{N}_2,\, \mathcal{N}_1(\sigma)). \tag{3}$$

(iii) Normalization: $R(\mathrm{id}, \sigma) = \mathrm{id}$.

(iv) Classical limit: *On commutative subalgebras,* R *reduces to classical Bayes' rule.*

Then, for faithful σ*,* $R(\mathcal{N}, \sigma) = \mathcal{R}_{\sigma,\mathcal{N}}$ *is the unique such functor.*

Proof sketch.—Axioms (ii) and (iv) force the $\sigma^{1/2}(\cdot)\sigma^{1/2}$ sandwich structure. For commutative algebras, R must be the Bayesian inverse $R(N,p)(q)_i = p_i N_{j|i}/(\sum_k p_k N_{j|k})$, which in matrix form is $p^{1/2} N^T (Np)^{-1/2}(\cdot)(Np)^{-1/2} N p^{1/2}$. By the functoriality axiom (3) and the Stinespring dilation theorem, this extends uniquely to the noncommutative setting as Eq. (2). The full proof is in Ref. [6]; see also [9] for the Bayesian framework and Supplemental Material [34] for a self-contained derivation. □

Quantum eraser as Petz recovery.—In the quantum eraser, a signal-idler pair is prepared in the entangled state $|\psi\rangle_{SE} = (|A, d_A\rangle + |B, d_B\rangle)/\sqrt{2}$, where $|A\rangle, |B\rangle$ are path states and $|d_A\rangle, |d_B\rangle$ are orthogonal detector states. The which-path channel is $\mathcal{N} = \mathrm{Tr}_E$, giving the fully decohered signal state $\rho_S = \frac{1}{2}(|A\rangle\langle A| + |B\rangle\langle B|)$. To our knowledge, the following explicit identification has not appeared previously (concurrent work by Bai, Buscemi, and Scarani [31] develops related ideas in the stochastic entropy production setting). Applying the Petz recovery (2) with $\sigma = |\psi\rangle\langle\psi|_{SE}$ and $\mathcal{N} = \mathrm{Tr}_E$:

$$\mathcal{R}_{\sigma,\mathrm{Tr}_E}(\rho_S) = |\psi\rangle\langle\psi|_{SE}, \tag{4}$$

which is exact because Tr_E saturates the DPI for pure bipartite states [34]. The "erasure measurement" on the idler (projecting onto $(|d_A\rangle \pm |d_B\rangle)/\sqrt{2}$) is the physical implementation of this Petz recovery. Strictly, the idler measurement is a *quantum instrument* (a collection of CP maps summing to a CPTP map); conditioned on outcome $\pm$, the effective map on the signal is CP but not trace-preserving. The Petz map $\mathcal{R}_{\sigma,\mathrm{Tr}_E}$ itself is CPTP and recovers the full entangled state (4); the subsequent projection onto $|\pm\rangle$ selects the subensemble where the coherent superposition $|A\rangle \pm |B\rangle$ is restored. The functoriality axiom (3) corresponds to the sequential structure of the eraser: composing the which-path marking with the idler measurement decomposes as retrodiction from the idler, then from the path.

FROM RETRODICTION TO ERROR CORRECTION

The uniqueness of the Petz map as retrodiction functor connects naturally to its role as a QEC decoder. The following collects known results; our contribution is the interpretive link.

Theorem 2 (Near-Optimal QEC via Petz Recovery (Barnum–Knill [5], Junge et al. [11]))**.** *Let* $\mathcal{N}$ *be a noise channel acting on a* $[\![n, k, d]\!]$ *stabilizer code* $\mathcal{C}$ *with code projector* P *and maximally mixed code state* $\sigma_{\mathcal{C}} = P/\dim\mathcal{C}$*. Let* $\epsilon_{\mathrm{KL}} = \max_{a,b} \left\| P E_a^\dagger E_b P - c_{ab} P \right\|$ *measure deviation from the Knill–Laflamme conditions [10], where* $\{E_a\}$ *are Kraus operators of* $\mathcal{N}$*. Then:*

(a) *If* $\epsilon_{\mathrm{KL}} = 0$ *(exact QEC),* $\mathcal{R}_{\sigma_{\mathcal{C}},\mathcal{N}} \circ \mathcal{N} = \mathrm{id}$ *on* $\mathcal{C}$*.*

(b) *In general, the universal recovery theorem of Junge et al. [11] (Corollary 5.1 therein) guarantees the existence of a* rotated *Petz map* $\tilde{\mathcal{R}}_{\sigma,\mathcal{N}}$ *such that:*

$$F\big(\rho,\, \tilde{\mathcal{R}}_{\sigma_{\mathcal{C}},\mathcal{N}}(\mathcal{N}(\rho))\big)^2 \geq \exp\big(-\Delta D\big), \tag{5}$$

where $\Delta D = D(\rho\|\sigma_{\mathcal{C}}) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma_{\mathcal{C}}))$ *is the relative entropy decrease. The standard Petz map satisfies the weaker Barnum–Knill bound [5]:* $F_e \geq \frac{1}{2} F_{e,\mathrm{opt}}^2$*.*

Proof.—(a) When the Knill–Laflamme conditions hold exactly, the DPI is saturated for all code states: $D(\rho\|\sigma_{\mathcal{C}}) = D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma_{\mathcal{C}}))$ for all ρ supported on $\mathcal{C}$. By Petz's sufficiency theorem [4], the Petz recovery is exact. (b) The rotated Petz map is $\tilde{\mathcal{R}}(\cdot) = \int dt\; p(t)\, \sigma^{it} \mathcal{R}_{\sigma,\mathcal{N}}(\mathcal{N}(\sigma)^{-it}(\cdot)\mathcal{N}(\sigma)^{it})\sigma^{-it}$, where $p(t) = (\pi/2)(\cosh(\pi t) + 1)^{-1}$; the bound follows from Ref. [11]. □

QEC decoding is therefore retrodiction: the decoder retrodicts the original ρ from the corrupted $\mathcal{N}(\rho)$ using the code state as prior. The Petz map is the unique Bayesian-consistent choice (Theorem 1), and its near-optimality [5] can be understood as a consequence of this uniqueness.

THE THERMODYNAMIC STRUCTURE

When the reference state σ is a fixed point of the channel (e.g., a Gibbs state at thermal equilibrium), known results combine into a chain linking recovery fidelity to entropy production. We state this chain to establish the connection to the temporal asymmetry parameter τ.

Proposition (Master Inequality Chain (assembling [11–13, 18]))**.** *For any quantum channel* $\mathcal{N}$*, reference state* σ*, and state* ρ*:*

$$\boxed{-\log F^2 \;\leq\; I(A; E|B)_\omega \;\leq\; \Sigma \;\leq\; \Delta D} \tag{6}$$

where $F = F(\rho,\, \mathcal{R}_{\sigma,\mathcal{N}}(\mathcal{N}(\rho)))$ *is the recovery fidelity;* $I(A; E|B)_\omega$ *is the conditional mutual information of the Stinespring complementary output;* Σ *is the entropy production[?]; and* $\Delta D = D(\rho\|\sigma) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma))$ *is the*

relative entropy decrease. When $\sigma = e^{-\beta H}/Z$ is a Gibbs state:

$$\Delta D = \beta\, W_{\mathrm{diss}} = \Delta S_{\mathrm{irr}}/k_{\mathrm{B}}, \tag{7}$$

yielding a thermodynamic bound on recovery: $F^2 \geq e^{-\Delta S_{\mathrm{irr}}/k_{\mathrm{B}}}$.

Proof.—Here F denotes the fidelity with the rotated Petz map $\tilde{\mathcal{R}}_{\sigma,\mathcal{N}}$ of Ref. [11].

Step 1 (Fidelity bound): By the universal recovery theorem [11], $F^2 \geq \exp(-\Delta D)$, so $-\log F^2 \leq \Delta D$.

Step 2 (CMI bound): By Fawzi–Renner [12], $I(A;E|B) \geq -\log F^2$ via the strengthened data processing inequality applied to the Stinespring dilation, giving the leftmost bound.

Step 3 (Entropy production): For a thermodynamic process with Gibbs reference state $\sigma = e^{-\beta H}/Z$, the entropy production satisfies $\Sigma \geq I(A;E|B)$ because the Landauer component $\Sigma_{\mathrm{Landauer}} = D(\rho\|\sigma) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma))$ equals $\Delta D \geq I(A;E|B)$ (by Steps 1–2), and $\Sigma \geq \Sigma_{\mathrm{Landauer}}$ by the Kolchinsky–Wolpert decomposition [13].

Step 4 (Second law): The relative entropy decrease $\Delta D \geq \Sigma$ by Reeb–Wolf [18]: the free-energy dissipation exceeds the entropy production by system-bath correlations and bath displacement.

The tightness conditions are physically transparent: $F^2 = 1$ iff exact QEC (Knill–Laflamme satisfied); $I(A;E|B) = 0$ iff Markov condition (no information leaks to environment); $\Sigma = 0$ iff thermodynamically reversible process. See Supplemental Material [34] for the detailed derivation including the Gibbs specialization (7). □

Quantum Crooks connection.—When $\sigma = \sigma_\beta = e^{-\beta H}/Z$, the Petz map becomes $\mathcal{R}_{\sigma_\beta,\mathcal{N}}$ with the structural factor $\sigma_\beta^{1/2} = e^{-\beta H/2}/\sqrt{Z}$ —exactly the "half Boltzmann weight" appearing in the KMS condition of modular theory. Kwon and Kim [7] proved that this $\mathcal{R}_{\sigma_\beta,\mathcal{N}}$ is precisely the reverse channel in the quantum Crooks fluctuation theorem:

$$\frac{p_F(w)}{p_R(-w)} = e^{\beta(w-\Delta F)}, \tag{8}$$

where $p_F(w)$ and $p_R(-w)$ are forward and Petz-reversed work distributions. The detailed balance condition $\mathcal{R}_{\sigma,\mathcal{N}} = \mathcal{N}$ characterizes thermodynamically reversible processes—those that are their own Petz reversal.

Temporal asymmetry parameter.—The infidelity $1-F$ of the Petz recovery is a standard measure of how far a channel deviates from reversibility. We give it a physical name:

$$\tau \;=\; 1 - F\big(\rho,\, \tilde{\mathcal{R}}_{\sigma,\mathcal{N}}(\mathcal{N}(\rho))\big). \tag{9}$$

This parameter quantifies the degree to which the channel $\mathcal{N}$ distinguishes past from future when σ is the reference state. Two Petz maps appear in this work. The retrodiction uniqueness theorem [6] singles out the *standard* Petz map $\mathcal{R}_{\sigma,\mathcal{N}}$ as the unique retrodiction functor; the quantum Crooks theorem likewise uses $\mathcal{R}_{\sigma,\mathcal{N}}$ as the physical reverse channel. The bound (5) and the definition of τ require the *rotated* map $\tilde{\mathcal{R}}$, which averages $\mathcal{R}$ over modular automorphisms $\sigma^{it}(\cdot)\sigma^{-it}$. For a thermal reference $\sigma = e^{-\beta H}/Z$, the rotation $\sigma^{it} = e^{-it\beta H}/Z^{it}$ is simply unitary time evolution generated by H, so $\tilde{\mathcal{R}}$ smooths the standard Petz map over the system's own Hamiltonian dynamics. This thermal averaging handles quantum noncommutativity $[\rho,\sigma] \neq 0$ and ensures that the fidelity bound $F^2 \geq \exp(-\Delta D)$ holds universally. When $[\rho,\sigma] = 0$ (classical limit) or $\Delta D = 0$ (exact recovery), $\tilde{\mathcal{R}}$ reduces to $\mathcal{R}$ and the distinction vanishes. For a unitary channel $\mathcal{N} = \mathcal{U}$, the Petz map reduces to $\mathcal{U}^\dagger$ and $F = 1$, so $\tau = 0$: closed systems have no temporal asymmetry. More generally, $F^2 \geq \exp(-\Sigma)$ gives

$$\tau \;\leq\; 1 - \exp(-\Sigma/2), \tag{10}$$

which enforces $\tau = 0$ whenever $\Sigma = 0$. The converse also holds. For faithful σ (the Crooks link requires Gibbs σ; see [34], Sec. S11):

$$\tau = 0 \iff \Sigma = 0 \iff I(A;E|B) = 0 \iff \mathrm{QMC}, \tag{11}$$

where QMC denotes the quantum Markov chain condition [12]. Each individual equivalence is established [4, 12, 29]; the full chain (11) states them as a single result. The arrow of time ($\tau > 0$) is equivalent to environmental coupling ($I(A;E|B) > 0$), entropy production ($\Sigma > 0$), and imperfect recoverability ($F < 1$).

The quantum eraser illustrates this directly. The signal-idler pair is a closed bipartite system with $\mathcal{N} = \mathrm{Tr}_E$ and $\sigma = |\psi\rangle\langle\psi|_{SE}$. Since $I(A;E|B) = 0$ and $\Sigma = 0$, the equivalence chain gives $\tau = 0$: the Petz map achieves exact recovery (Sec.). By contrapositive, recovery of interference implies prior superposition [34]. In Wheeler's delayed-choice experiment [15], $\tau = 0$ implies the temporal order of measurements is irrelevant [27]: the "need to wait" for the idler measurement arises only when one traces out the idler ($\tau > 0$).

NEW RESULT I: BOUND SATURATION THEOREM

We characterize exactly when the Junge et al. bound $F^2 \geq \exp(-\Delta D)$ is saturated. The answer connects Petz recovery to the classical theory of sufficient statistics.

Theorem 3 (Saturation of the Recovery Bound)**.** *Let $\mathcal{N}$ be a CPTP map with $\sigma = I/d$ and $\rho = |\psi\rangle\langle\psi|$ a pure state. Write $\omega = \mathcal{N}(\rho)$ and $\tau = \mathcal{N}(\sigma)$. Then*

$$F^2\big(\rho,\, \mathcal{R}_{\sigma,\mathcal{N}}(\mathcal{N}(\rho))\big) = \exp(-\Delta D) \tag{12}$$

*if and only if[**?**]:*

(i) $[\omega, \tau] = 0$ (channel outputs commute);

(ii) $\omega_i/\tau_i = c$ *for all* i *with* $\omega_i > 0$ (constant likelihood ratio on support).

When these hold, $F^2 = c/d$ *and* $\Delta D = \ln(d/c)$*, where* $c = \omega_i/\tau_i$ *is the common ratio.*

Proof.—Sufficiency: When (i)–(ii) hold, $\tau^{-1/2}\omega\,\tau^{-1/2} = c\,P_{\mathrm{supp}(\omega)}$. Since each Kraus vector $E_k|\psi\rangle$ lies in the range of ω, we have $P_{\mathrm{supp}(\omega)}E_k|\psi\rangle = E_k|\psi\rangle$ for all k. Hence $F^2 = \frac{1}{d}\langle\psi|\mathcal{N}^\dagger(c\,P_{\mathrm{supp}(\omega)})|\psi\rangle = \frac{c}{d}\sum_k \|E_k|\psi\rangle\|^2 = c/d$. Meanwhile $D(\omega\|\tau) = \sum_i \omega_i \ln(c) = \ln c$, so $\exp(-\Delta D) = \exp(\ln c - \ln d) = c/d$.

Necessity: When (i) holds, in the joint eigenbasis of ω and τ, $F^2 = \frac{1}{d}\sum_i \omega_i^2/\tau_i$ (weighted arithmetic mean of $x_i = \omega_i/\tau_i$) and $\exp(-\Delta D) = \frac{1}{d}\prod_i x_i^{\omega_i}$ (weighted geometric mean). By the weighted AM–GM inequality, $F^2 \geq \exp(-\Delta D)$ with equality if and only if all x_i are equal on the support—i.e., condition (ii). When (i) fails ($[\omega, \tau] \neq 0$), strict inequality $F^2 > \exp(-\Delta D)$ is verified numerically for 5×10^5 random non-commuting pairs with zero exceptions [34]. □

Sufficient-statistic interpretation.—In classical statistics, a statistic T is *sufficient* for discriminating P from Q iff the likelihood ratio P/Q is constant on each level set of T. Conditions (i)–(ii) are the quantum analogue: the channel $\mathcal{N}$ acts as a quantum sufficient statistic for the pair $\{\rho, \sigma\}$, preserving the distinguishability ratio uniformly on the support. Saturation of the Petz bound is equivalent to sufficiency of the channel.

Examples.—(a) Amplitude damping $\mathcal{N}_\gamma^{\mathrm{AD}}$, $\rho = |0\rangle\langle 0|$: $\omega = |0\rangle\langle 0|$ (pure), $\tau = ((1+\gamma)|0\rangle\langle 0| + (1-\gamma)|1\rangle\langle 1|)/2$, ratio $c = 2/(1+\gamma)$, $F^2 = 1/(1+\gamma)$. Saturated. (b) Completely depolarizing $\mathcal{N}(\rho) = I/d$: $\omega = \tau = I/d$, ratio $c = 1$, $F^2 = 1/d$. Saturated for *all* pure inputs. (c) Amplitude damping, $\rho = |1\rangle\langle 1|$: ratios $\omega_0/\tau_0 \neq \omega_1/\tau_1$. Not saturated. (d) Depolarizing $0 < p < 1$: $\omega \neq \tau$, ratios unequal. Not saturated.

Experimental relevance.—Amplitude damping (T_1 decay) is the dominant noise in superconducting qubits and ion traps, and Petz recovery has been implemented on both platforms [32, 33]. The prediction $F^2 = 1/(1+\gamma)$ for ground-state input is directly testable.

NEW RESULT II: COMPOSITIONAL SUB-ADDITIVITY OF τ

The Petz map's functoriality (3) implies a geometric structure on τ that other recovery maps lack.

Theorem 4 (Composition Sub-additivity)**.** *For sequential channels* $\mathcal{N}_1, \mathcal{N}_2$ *and reference state* σ*, the Petz recovery* τ *satisfies:*

$$\sqrt{\tau_{12}} \;\leq\; \sqrt{\tau_1} + \sqrt{\tau_2^{\mathrm{eff}}}, \tag{13}$$

where $\tau_{12} = 1 - F(\rho,\, \mathcal{R}_{\sigma,\mathcal{N}_2\circ\mathcal{N}_1} \circ (\mathcal{N}_2\circ\mathcal{N}_1)(\rho))$*,* $\tau_1 = 1 - F(\rho,\, \mathcal{R}_{\sigma,\mathcal{N}_1}\circ\mathcal{N}_1(\rho))$*, and* $\tau_2^{\mathrm{eff}} = 1 - F(\mathcal{N}_1(\rho),\, \mathcal{R}_{\mathcal{N}_1(\sigma),\mathcal{N}_2}\circ \mathcal{N}_2(\mathcal{N}_1(\rho)))$ *is the second-stage cost evaluated at the* evolved *state* $\mathcal{N}_1(\rho)$ *with* updated *reference* $\mathcal{N}_1(\sigma)$*.*

Proof.—Three ingredients. (1) *Functoriality* (3): $\mathcal{R}_{\sigma,\mathcal{N}_2\circ\mathcal{N}_1} = \mathcal{R}_{\sigma,\mathcal{N}_1} \circ \mathcal{R}_{\mathcal{N}_1(\sigma),\mathcal{N}_2}$. (2) *Bures triangle inequality*: for any three states ρ, ρ_1, ρ_{12}, $\sqrt{1 - F(\rho, \rho_{12})} \leq \sqrt{1 - F(\rho, \rho_1)} + \sqrt{1 - F(\rho_1, \rho_{12})}$. (3) *Data-processing inequality*: $\mathcal{R}_{\sigma,\mathcal{N}_1}$ is CPTP, so $F(\rho_1, \rho_{12}) = F(\mathcal{R}_{\sigma,\mathcal{N}_1} \circ \mathcal{R}_{\mathcal{N}_1(\sigma),\mathcal{N}_2} \circ \mathcal{N}_2 \circ \mathcal{N}_1(\rho),\, \mathcal{R}_{\sigma,\mathcal{N}_1} \circ \mathcal{N}_1(\rho)) \geq F(\mathcal{R}_{\mathcal{N}_1(\sigma),\mathcal{N}_2} \circ \mathcal{N}_2(\mathcal{N}_1(\rho)),\, \mathcal{N}_1(\rho))$. Combining gives (13). See [34], Sec. S18. □

Remark.—The reference for the second stage *must* be the Bayesian-updated $\mathcal{N}_1(\sigma)$, not the original σ. With the original reference, the inequality fails numerically. This is physically meaningful: irreversibility decomposes additively across sequential processes only when each stage's cost is measured using the updated prior—the hallmark of Bayesian inference.

Uniqueness to Petz.—The rotated Petz map $\tilde{\mathcal{R}}$ of Ref. [11], which gives tighter fidelity bounds, does *not* satisfy Eq. (13): numerical tests find violations for non-unital channel pairs (see [34], Sec. S18). This provides a new structural distinction between the two recovery maps beyond what the categorical axioms [6] alone imply. Note that Theorem 4 uses the *standard* Petz map, while the bound (10) uses the *rotated* map. For commuting $[\rho, \sigma] = 0$ the two maps coincide; in general the rotated map achieves higher fidelity, so $\tau_{\mathrm{std}} \geq \tau_{\mathrm{rot}}$.

OBSERVATIONS FROM THE τ FRAMEWORK

Two further observations follow from interpreting existing results through τ.

Observation 1: Post-selection as thermodynamic filtering.—For a stabilizer code under noise $\mathcal{N}$, each syndrome s has a data-processing gap $\Delta D(s)$. Since $\Delta D(s)$ depends only on the noise channel and code state, the per-distance scaling of rejection rates should be decoder-independent. Numerical simulations at $d \in \{3, 5, 7, 9, 11\}$ with three decoders yield a coefficient of variation CV = 0.091 across decoders, consistent with this expectation [34]. This is also consistent with the Ising–QEC mapping [21], statistical-mechanical thresholds [20], and scalable post-selection results [19, 22].

Observation 2: Decoder hierarchy as retrodiction approximation.—Since the Petz map is the unique Bayesian-consistent retrodiction (Theorem 1), all good decoders must approximate it. The known decoder hierarchy [23]—ML > neural network [24] > MWPM > Union-Find—is consistent with being ordered by retrodiction fidelity. A direct test would require computing the channel divergence $D(\tilde{\mathcal{R}}_{\sigma,\mathcal{N}}\|\mathcal{D})$ for each decoder $\mathcal{D}$, which we leave to future work.

DISCUSSION

The parameter $\tau = 1 - F$ reads the Petz recovery fidelity as a continuous measure of temporal asymmetry: $\tau = 0$ for closed systems, $\tau > 0$ whenever the process is irreversible. The equivalence chain (11) makes this precise by connecting $\tau = 0$ to vanishing entropy production, the quantum Markov condition, and detailed balance. Beyond this interpretive framework, Theorem 3 gives a complete characterization of when the recovery bound saturates—the quantum analogue of a sufficient statistic—and Theorem 4 shows that $\sqrt{\tau}$ satisfies a triangle inequality unique to the standard Petz map.

The decoherence program [27, 28] provides the mechanism: environment-induced decoherence produces entropy ($\Sigma > 0$), forcing $\tau > 0$. The τ framework restates this in the Petz language, adds the saturation and composition structure (Theorems 3–4), and makes the connection to QEC decoders explicit.

Theorem 3 is proved for $\sigma = I/d$; the extension to general full-rank σ remains open, as does the continuous-time generalization of Theorem 4. For non-Markovian processes, Σ can be temporarily negative [30], implying transient decreases in τ. Whether the process tensor framework [16] extends the equivalence chain (11) to this setting is an important open question, as is a possible holographic interpretation of τ via the Petz map's role in entanglement wedge reconstruction [8].

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[34] See Supplemental Material for detailed derivations, the explicit quantum eraser calculation, the bound saturation if-and-only-if proof (Sec. S17), the composition subadditivity proof (Sec. S18), post-selection analysis, the equivalence chain proof, and caveats.